Transient Heat Conduction in Large Plane Walls, Long Cylinders, and Spheres

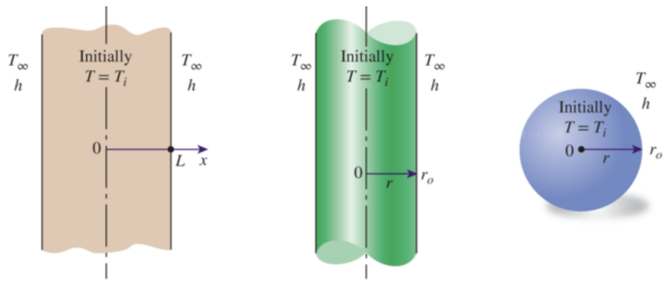
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Introduction

Transient heat conduction occurs when the temperature within a solid changes with both space and time. Unlike lumped analysis, the solid is not assumed to remain at one uniform temperature. Instead, temperature gradients start to develop inside the body.

The poster presents transient one-dimensional heat conduction in three common geometries and outlines the mathematical derivation using separation of variables and Sturm-Liouville theory.



(a) A large plane wall (b) A long cylinder (c) A sphere

Figure 1. One-dimensional transient heat conduction geometries [1].

Assumptions

The transient conduction analysis is based on the following assumptions:

1. Heat conduction is transient
2. No internal heat generation
3. Heat conduction is 1-D
4. Thermal properties are constant
5. The convection coefficient h is constant
6. Radiation is neglected

Governing equation: Diffusion equation (plane wall)

The transient heat conduction in a plane wall is governed by

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}, \qquad \text{(Heat)}$$

where $\alpha = \frac{k}{\rho C_p}$, is the thermal diffusivity, k is the thermal conductivity, ρ is the density, and C_p is the specific heat. For this problem, the initial and symmetry conditions are

$$T(x,0) = T_i, \quad \frac{\partial T}{\partial x}(0,t) = 0.$$

These conditions describe a solid that is initially at uniform temperature T_i and is symmetric about its centerline.

In transient conduction, the surface temperature changes first when the body is exposed to a fluid at temperature T_∞. Because the interior does not respond instantly, a temperature gradient develops inside the solid.

This is why transient conduction problems must account for both time and position.

Nondimensionalization

Nondimensionalization transforms the governing equations into a form that is independent of specific material properties and geometry

This is useful because:

- It reduces the number of variables in the problem
- It allows different physical systems to be compared using the same solution
- All materials follow the same dimensionless equation after scaling

The *dimensionless temperature* is defined as: $\Theta(x,t) = \dfrac{T(x,t) - T_\infty}{T_i - T_\infty}$ and the *dimensionless position* and time are:

$$X = \frac{x}{L}, \qquad \tau = \frac{\alpha t}{L^2}$$

The Biot number is: $\mathrm{Bi} = \dfrac{hL}{k}$, where h is the convection coefficient, L is the characteristic length, and k is the thermal conductivity. The Biot number compares internal heat conduction resistance to surface convection resistance.
After nondimensionalization, the problem is governed primarily by the Fourier number and Biot number.

Dimensionless Equation and Boundary Conditions

The dimensionless heat equation is

$$\frac{\partial \Theta}{\partial \tau} = \frac{\partial^2 \Theta}{\partial X^2}, \quad \Theta(X,0) = 1, \quad \Theta_X(0,\tau) = 0, \quad \Theta_X(1,\tau) = -\mathrm{Bi}\,\Theta(1,\tau).$$

This is a Robin (convection) BCs at $X = 1$.

Separation of variables:

$$\Theta(X,\tau) = F(X)G(\tau), \qquad \frac{G'}{G} = \frac{F''}{F} = -\lambda.$$

$$G' + \lambda^2 G = 0, \qquad F'' + \lambda^2 F = 0.$$

Sturm–Liouville problem:

$$F'' + \lambda^2 F = 0, \quad F'(0) = 0, \quad F'(1) = -\mathrm{Bi} \cdot F(1).$$

General solution:

$$F(X) = C_1 \cos(\lambda X) + C_2 \sin(\lambda X).$$

From symmetry:

$$F'(0) = 0 \Rightarrow C_2 = 0, \qquad F(X) = C\cos(\lambda X).$$

At $X = 1$:

$$-C\lambda \sin(\lambda) = -\mathrm{Bi} \cdot C\cos(\lambda) \;\Rightarrow\; -\lambda \sin(\lambda) = -\mathrm{Bi} \cdot \cos(\lambda) \;\Rightarrow\; \boxed{\lambda \tan(\lambda) = \mathrm{Bi}}.$$

Eigenfunctions:

$$F_n(X) = \cos(\lambda_n X), \qquad \lambda_n \tan(\lambda_n) = \mathrm{Bi}.$$

Time solution:

$$G' + \lambda_n^2 G = 0 \;\Rightarrow\; G_n(\tau) = e^{-\lambda_n^2 \tau}.$$

Series solution:

$$\Theta(X,\tau) = \sum_{n=1}^{\infty} A_n \cos(\lambda_n X) e^{-\lambda_n^2 \tau}.$$

This shows that the temperature distribution is represented by a cosine series in space with exponential decay in time.

Temperature Distribution in Space and Time (ANSYS)

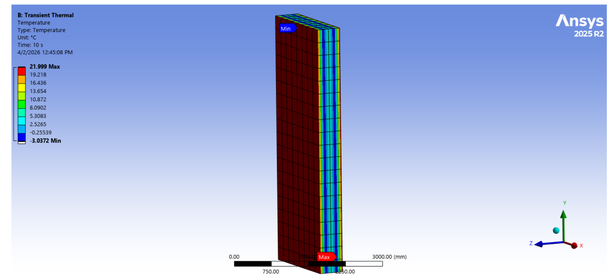


Figure 2. Temperature Distribution in Space and Time (ANSYS)

Transient thermal simulation of a plane wall showing temperature as a function of time and space. The smooth transition from hot to cold regions across the wall agrees with the heat equation solution.

Analytical Solution by Separation of Variables

Geometry	Solution	Eigenvalue condition
Plane wall	$\Theta = \sum_{n=1}^{\infty} \frac{4\sin\lambda_n}{2\lambda_n + \sin(2\lambda_n)} e^{-\lambda_n^2 \tau} \cos(\lambda_n x/L)$	$\lambda_n \tan(\lambda_n) = \mathrm{Bi}$
Cylinder	$\Theta = \sum_{n=1}^{\infty} \frac{2}{\lambda_n} \frac{J_1(\lambda_n)}{J_0^2(\lambda_n) + J_1^2(\lambda_n)} e^{-\lambda_n^2 \tau} J_0(\lambda_n r/r_0)$	$\frac{J_1(\lambda_n)}{J_0(\lambda_n)} = \mathrm{Bi}$
Sphere	$\Theta = \sum_{n=1}^{\infty} \frac{4(\sin\lambda_n - \lambda_n \cos\lambda_n)}{2\lambda_n - \sin(2\lambda_n)} e^{-\lambda_n^2 \tau} \frac{\sin(\lambda_n r/r_0)}{\lambda_n r/r_0}$	$1 - \lambda_n \cot(\lambda_n) = \mathrm{Bi}$

Table 1. General solutions for plane walls, cylinders, and spheres [1]

One-Term Approximation and Common Geometries

For $\tau > 0.2$, the first term of the series provides a good approximation, with errors less than 2%.

$$\Theta \approx A_1 e^{-\lambda_1^2 \tau} \cos(\lambda_1 X)$$

Similar approximations apply to cylinders and spheres.

The constants A_1 and λ_1 depend on the Biot number and can be obtained from standard tables or Heisler charts.

Future Work

Future work will extend the model by incorporating radiation effects and a time-dependent convection coefficient $h(t)$ in the boundary conditions. The resulting nonlinear boundary behavior motivates the use and development of numerical methods for analysis and simulation.

References

[1] Yunus A. Cengel and Afshin J. Ghajar. *Heat and mass transfer: Fundamentals and applications*. McGraw-Hill Education, 2025.

[2] W. A. Strauss. *Partial differential equations*. John Wiley & Sons, Ltd., Chichester, second edition, 2008. An introduction.

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